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# Impact of the MW-M31 Merger on the Dark Matter Halo Distribution of M33

Nicolas Mazziotti[1]

[1] *University of Arizona*

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## 1. INTRODUCTION

The majority of a galaxy's mass is believed to come from its **dark matter halo**, a cloud-like structure of non-luminous matter that envelops the visible components of a galaxy. Existence of the dark matter halo was first hypothesized to explain the unusually high rotation speeds in the outskirts of nearby spiral galaxies, implying that there was significantly more matter present in these galaxies than could be observationally detected (F. Zwicky 1933; V. C. Rubin & W. K. Ford 1970). Although dark matter particles are invisible and weakly-interacting, the three dimensional distribution of the dark matter halo around a galaxy—known as the **halo shape**—is not considered to be random. Geometrical conventions such as **oblate** (elongated along poles), **prolate** (flattened at poles), or **triaxial** (asymmetric poles) matter distributions are commonly used metrics for quantifying the halo shape, based off of analyses of observational and simulated data.

Dark matter and baryons are the fundamental building blocks of a **galaxy**. Relating the dark matter halo shape to observable gas and stellar features in galaxies across cosmic time is essential for studying **galaxy evolution**, which correlates the morphology of a galaxy to a specific evolutionary stage. This paradigm forms the basis of the Lambda **cold dark matter (ΛCDM) theory** of cosmology, which asserts that dense regions of dark matter in the early universe provided the necessary scaffolding for galaxy formation by pulling in sufficient amounts of baryonic matter. Galaxies in the **Local Group**, namely the Milky Way (MW), M31, and M33, are ideal candidates to study dark matter halos because of their proximity, especially the halos of satellites that would otherwise be too distant to probe. This may lend valuable insight into a longstanding issue in ΛCDM cosmology: the missing satellites problem, which arises from the fact that fewer satellites have been found in the Local Group than predicted (G. Kauffmann et al. 1993).

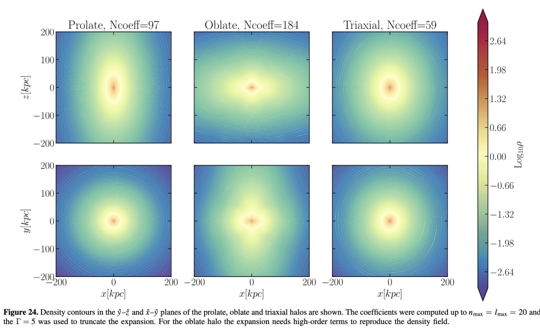


**Figure 1.** Figure 24 taken from N. Garavito-Camargo et al. (2021) displaying the morphology of prolate (left column), oblate (middle column), and triaxial (right column) dark matter halos. The top row is the x-z projection while the bottom row is x-y.

Efforts to characterize the 3D shape of dark matter halos are largely driven by simulations, due to the invisible nature of dark matter. By examining Milky Way sized dark matter halos in the Aquarius cosmological N-body simulations, C. A. Vera-Ciro et al. (2011) found that halo shape measured at the virial radius typically evolved from prolate to a more triaxial/oblate distribution. Narrow filaments of infalling material were found to correlate more with prolate configurations, while isotropic accretion gave rise to triaxial/oblate configurations. However, galaxy interactions may cause significant deviations from these three shapes. In N. Garavito-Camargo et al. (2021), N-body simulations of the LMC passing through the Milky Way showed that asymmetric perturbations were induced in the MW halo that did not fit a prolate, oblate, or triaxial distribution. Idealized versions of these three halo shapes computed by N. Garavito-Camargo et al. (2021) are shown in Figure 1 for reference.

The connection between dark matter subhalos and the missing satellites problem is an area of ongoing research. A thorough exploration into the geometrical evolution of satellite sized halos, as opposed to Milky Way sized halos, has yet to be conducted, though simulations of galaxies in the Local Group have grown significantly in resolution. An interesting candidate is M33, as it is the third largest galaxy in the Local Group and lacks a central bulge common for spiral galaxies. Investigating how the shape of M33's dark matter halo evolves as it orbits M31 and the eventual merger between M31 and the MW could probe the stability of such satellite galaxies. A significant shift in halo configuration could be indicative of a process that might have destabilized many satellites in the early universe.

## 2. THIS PROJECT

In this paper, I aim to utilize the simulation data provided by R. P. van der Marel et al. (2012) of the MW-M31-M33 system to characterize the shape of M33's dark matter halo from the present day until ∼12 Gyr in the future. I will attempt to characterize the 3D halo shape of M33 by examining isodensity contours of the dark matter halo distribution in different 2D projections.

This paper seeks to address how the halo shape is impacted by the MW-M31 merger in ∼4 Gyr, if at all, and how this agrees with the predictions made by C. A. Vera-Ciro et al. (2011) about the evolution of halo shape. If the distribution becomes more triaxial/oblate post-merger, then the accretion of mass onto M33 as a result of the merger may have been more isotropic. If the distribution instead becomes more prolate, then narrow filaments of mass accretion may be to blame. However, if the distribution remains the same, then a massive galaxy merger such as this may not have much of an effect on the halo shape of satellites.

Investigating how the halo shape of a M33-class satellite changes in response to a galaxy merger is important for understanding how dark matter halos evolved early in the universe when galaxy mergers were commonplace for large-scale structure formation. As a larger than average satellite, any significant changes to the halo shape of M33 can be generalized to lower mass satellites, helping to better understand satellite populations and their evolution.

## 3. METHODOLOGY

I rely on the data from R. P. van der Marel et al. (2012) produced using collisionless N-body simulations of the MW-M31-M33 system, which models the dynamics of particles belonging to the dark matter halo, disk, and bulge components of these galaxies. The gaseous component is not included in the simulations because of its substantially low fraction compared to the stellar mass in these galaxies. The dark matter halos are initially modeled with a Hernquist (1990) profile.

Most of my methods are taken from Lab 7. I plan to use the high resolution dark matter particle data of M33 in the simulation (particle type 1) to extract information about M33's halo. First, I will use the `CenterOfMass` code to calculate the center of mass of M33 according to the dark matter particles at the desired snapshot. The dark matter particles are used to calculate the COM as opposed to the disk particles because the halo COM will likely differ from the disk particle COM. I can then calculate positions of all of the dark matter particles in the COM frame. I will then use the `RotateFrame` function to align the particle coordinates frame such that the x-y projection is a face-on view of M33, and the x-z and y-z projections are edge-on. These coordinates will be used to make a projected density plot as in Lab 7 using the `matplotlib` 2D histogram, overlayed with fitted elliptical contours produced with `photutils` (see Fig. 2). This plot should have dense contours like in Fig. 1 to see the halo shape more clearly. To define the field of view, I will use the radial distance at which the average DM density enclosed within a sphere of the halo is 500 times the critical density of the universe, $R_{500}$. The mass enclosed will be computed with the `massEnclosed` method inside the `MassProfile` class. The length of the projection box will therefore always be $2R_{500}$.

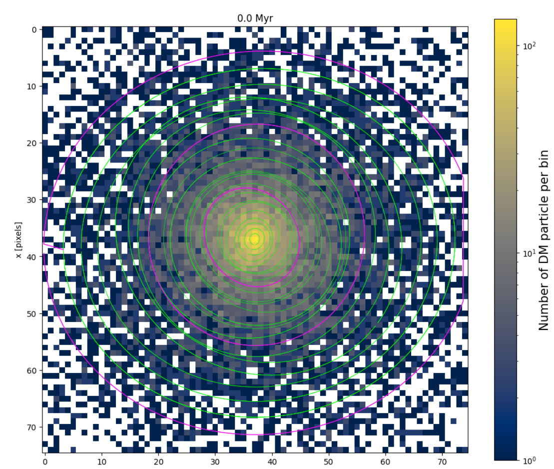


**Figure 2.** Face-on density projection (x-y) of the dark matter particles in M33 at Snapshot 0. Isodensity cntours are overlayed in green. The outer magenta line corresponds to roughly the halo $R_{500}$ radius, followed by the $R_{500}/2$ and $R_{500}/4$ radii.

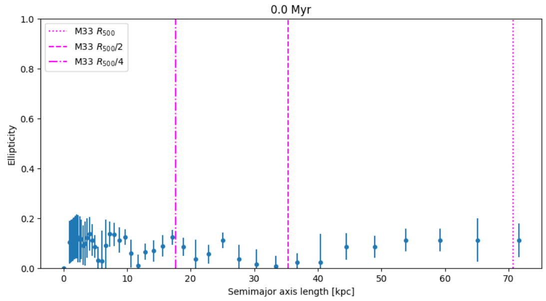


**Figure 3.** Corresponding ellipticity of the ellipses from Fig. 2 as a function of semimajor axis length in kpc. The dotted magenta line indicates the $R_{500}$ radius, the dashed line being the $R_{500}/2$ radius, and the semi-dotted line being the $R_{500}/4$ radius.

I will attempt to classify the halo shape as either prolate, oblate, or triaxial using these fitted ellipses. I will use the ellipticity at three representative distances in the halo as simple metrics to quantify the halo shape: $R_{500}$, $R_{500}/2$, and $R_{500}/4$. With the ellipticity $f$ at each of these distances, I can calculate the axis ratio $AR$ via:

$$AR = 1 - f \tag{1}$$

To quantify the halo shape, I will use the triaxiality paramater as defined by N. Garavito-Camargo et al. (2021):

$$T = \frac{1 - (b/a)^2}{1 - (c/a)^2} \tag{2}$$

where $a$, $b$, and $c$ are the major, intermediate, and minor axes of the 3D halo shape ($a > b > c$). A prolate shape is observed for $T \geq 0.67$, oblate for $T \leq 0.33$, and triaxial for the range in between. Perfect triaxiality is achieved at $T = 0.5$. By comparing the ellipticity in the x-y, y-z, and x-z projections, I will use the projection with the lowest ellipticity to compute a value for the $b/a$ axis ratio. The projection with the highest ellipticity will be used to compute the $c/a$ axis ratio. Even though these numbers may differ from the true axis ratios due to projection effects, they are a good enough ballpark estimate. I will attempt to compute a triaxiality value for the full snapshot range but am most interested in the present day, the end of the simulation, right before the first pericenter approach of M33 relative to M31, and then when the next apocenter is reached after first pericenter. I will use the orbital separation plots from Homework 6 to determine these precise snapshot numbers, which are around 4 Gyr in the future, the time of the MW-M31 merger. I will create a miniature movie of the density projections during the first pericenter-apocenter range to see how the halo shape visually changes.

I hypothesize that the halo distribution of M33 will be more triaxial at the present day because it has not been recently perturbed by M31 or any other object. As the M31-MW merger occurs, I predict that the halo shape of M33 will become more prolate due to the added mass of the Milky Way, elongating M33's halo along the collision axis. I don't think the halo will deviate from a prolate, oblate, or triaxial distribution post-merger because M33 is currently far enough away from M31 and massive enough to not be completely tidally disrupted by the merger. Additionally, dark matter does not interact with itself so the halo shape should remain approximately stable despite the introduction of the MW dark matter halo during the merger.

## 4. RESULTS

The halo triaxiality evolution of M33 is plotted in Figure 4, from the present day to ∼5 Gyr in the future. I was unable to fit ellipses beyond this due to irregularities in the density distribution, which also caused the fitting algorithm to fail at certain snapshots and create gaps in snapshot sampling. The triaxiality at the $R_{500}$, $R_{500}/2$, and $R_{500}/4$ distances are plotted as separate lines. Horizontal lines indicating $T = 0.33$ and 0.67 are shown for reference. At the present day, the halo shape appears to be more oblate except at $R_{500}/2$, where it is slightly triaxial. Up until 2 Gyr, the shape at all three distances fluctuates within the triaxial regime, occasionally becoming more prolate or oblate in some places. Beyond 2 Gyr, the triaxiality at these three distances generally follows the same trends: the halo is more oblate at 3.5 Gyr, after which it rapidly becomes more prolate by 4 Gyr, then drops down to being oblate again. The drastic changes in triaxiality at all three distances near the time of the MW-M31 merger at 4 Gyr suggest that the global halo shape is changing in response to the merger.

Figure 5 displays the evolution of the M33 DM halo from an face-on perspective (x-y projection) shortly before (left panel), at (middle panel), and after 4 Gyr (right panel). At 3.57 Gyr, the distribution looks somewhat elliptical, then is more elliptical at 4 Gyr and remains elliptical at 4.86 Gyr but with a more spread out distribution of matter. Vertical lines corresponding to these points in time are included in Figure 4. The left panel of Figure 5 therefore shows a more oblate shape before the merger and a more prolate shape around the time of the first pericenter approach between the MW and M31.

## 5. DISCUSSION

My analysis suggests that the DM halo shape of M33 at the present day is more oblate, then momentarily be-

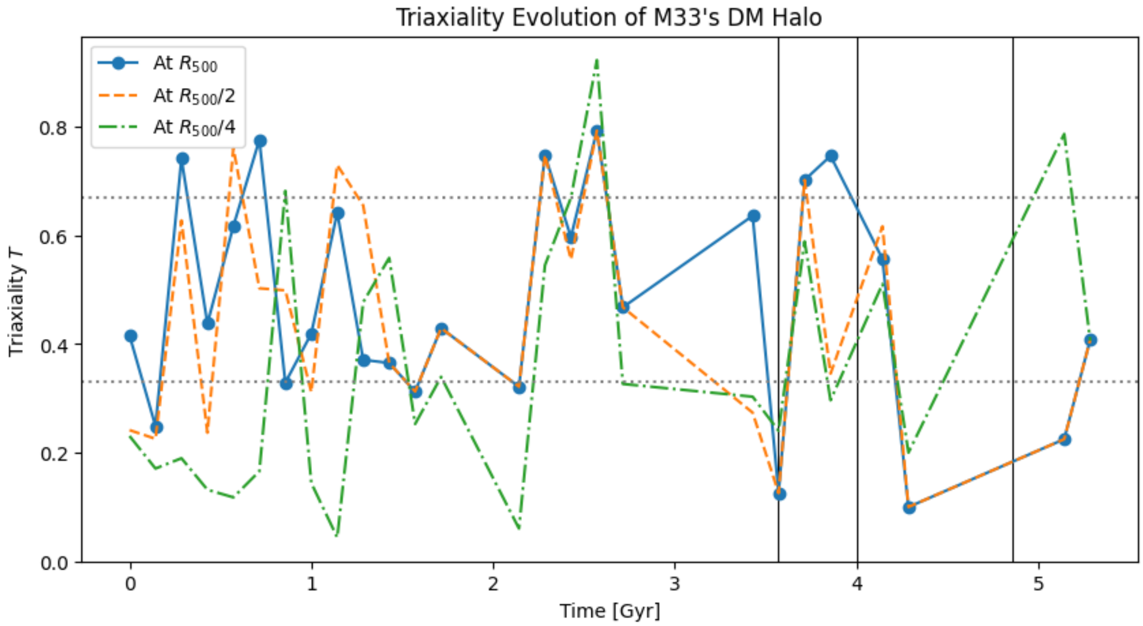


**Figure 4.** Computed triaxiality of the M33 DM halo at $R_{500}$, $R_{500}/2$, and $R_{500}/4$ as a function of time. The upper horizontal dotted line denotes $T = 0.67$ while the bottom one denotes $T = 0.33$. The vertical lines are added in reference to Figure 5. Dots are shown in the $R_{500}$ line to indicate the sampling rate of snapshots, where each dot is a snapshot where the ellipse fitting was successful across all three projection axes.

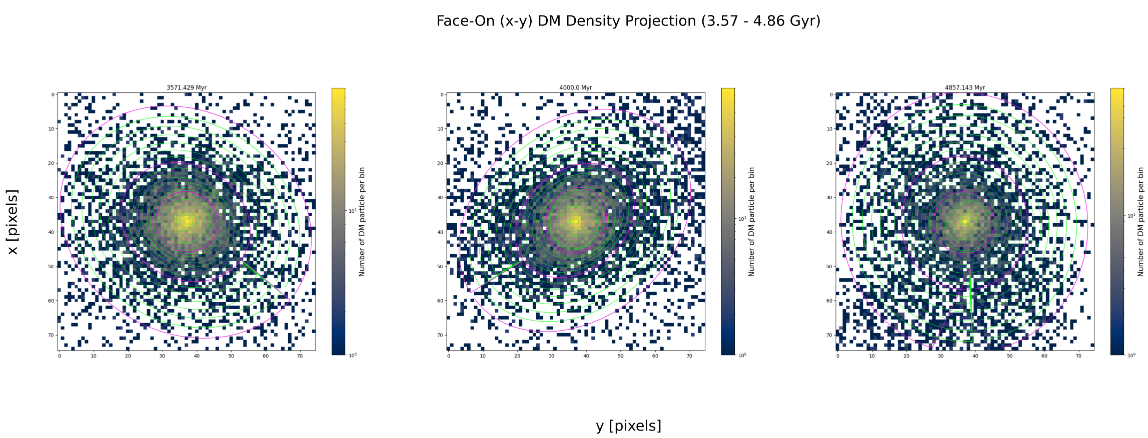


**Figure 5.** Density projections from a face-on perspective of the M33 DM halo at 3.57 Gyr (left panel), 4 Gyr (middle panel) and 4.86 Gyr (right panel). These snapshots were selected to show possible changes to the halo shape near the time of the MW-M31 merger.

comes more prolate around the beginning of the MW-M31 merger, after which it becomes oblate again. I hypothesized that the halo shape would be more triaxial at the present day, so this result does not agree with my prediction. The halo shape becoming more prolate in response to the MW-M31 merger, however, does agree with my hypothesis.

Using the mechanisms proposed by C. A. Vera-Ciro et al. (2011) that give rise to a prolate, oblate, or triaxial halo shape as a baseline, the sudden switch to a prolate configuration around the time of the MW-M31 merger may be because mass is being channeled into M33 along the direction in which the MW and M31 are approaching each other. The halo shape fluctuates between oblate and triaxial long before the merger possibly because M33 is accreting mass isotropically as it orbits M31. The transition from a prolate to oblate configuration after the merger begins suggests that mass accretion again becomes more isotropic once M31 and M33 pass by each other, possibly as a result of the time needed to spread around mass in all directions.

I limited my sampling in time to every 10 snapshots for computational efficiency but was unfortunately not always able to generate ellipticity values due to failures in the ellipse fitting process. Higher resolution sampling is needed to see if the trends I find in triaxiality are preserved or are more stochastic. A limitation of my methodology is the fact that the DM halo is not disk-shaped, so defining face-on and edge-on projection coordinate frames is not the most accurate. The projection where one would expect the ellipticity to be the highest is thus not always true. It might not be possible at times to measure the maximum ellipticity because of these projection effects, which in turn affects the measured triaxiality.

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